

HIRUY KASSA

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EDUCATION

University of St. Thomas

B.S. Computer Science

Sep. 2023 – Dec. 2027

GPA: 3.65 | Dean's List

Coursework: Data Structures, Object-Oriented Design, Computer Architecture, Information Security, Algorithms, Data Regression

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, Java, C++, Bash

Frameworks & Libraries: React, Flask, Node.js, Pandas, BeautifulSoup, pytest

DevOps & Cloud: Docker, GitHub Actions (CI/CD), AWS (EC2, VPC, IAM, CloudWatch, S3), Linux, Nginx

Tools: Git, GitHub, VS Code, Draw.io, TeamDynamix

PROJECTS

Ask the Early Church | React, Flask, Python, SQLite, Voyage AI, Docker, AWS

Mar. 2026 – Present

- Served **1,000+** daily active users by architecting a **hybrid search engine** over **52,000+** passages (**247 authors, 2,858 works**) fusing **Voyage AI vector embeddings** with FTS5 keyword search via reciprocal rank fusion and per-work/author diversification.
- Cut median search latency **~40%** by parallelizing **Voyage AI** embedding with **Gemini** query parsing in a worker thread and caching repeat queries (30-day TTL), keeping the corpus in RAM via a **float16 loader** (**~108MB** peak).
- Engineered a **5-stage ETL pipeline** (scrape, clean, chunk, embed, index) with BeautifulSoup, normalizing **~50K verse-level commentaries across 76 books** into a **6-table SQLite schema** powering a verse-first scripture browser.
- Shipped a **React + Vite** frontend (library across **5 categories** + book reader) on a **Flask REST API**, containerized with **Docker** on **AWS EC2** with **CI/CD via GitHub Actions**, **CloudWatch** monitoring, and a **2,997-URL** SEO sitemap.

FZ Contracting — AI Workflow System Design | Claude AI, Draw.io

Mar. 2026 – Present

- Designed a **6-stage agentic AI workflow** automating 5 manual business processes for a real contracting business; authored a **22-requirement PRD** with 4 architecture diagrams mapping API integrations (Google Calendar, Maps, Twilio SMS).

EXPERIENCE & LEADERSHIP

Technical Support

Mar. 2026 – Present

University of St. Thomas, ITS Desktop Services & Support

Minneapolis & St. Paul, MN

- Resolve **15–20 hardware, software, and network incidents per week** for **9,000+** users across Windows, Mac, and Linux environments, tracking all requests in **TeamDynamix**.
- Perform full-time summer deployment of **30+ new workstations per week**, handling OS install, software configuration, and domain join across a multi-campus environment.
- Manage **IT asset tracking** for hardware deployments by cleaning inventory data and logging device assignments.

IT Help Desk

Jan. 2025 – May 2025

St. Olaf College

Northfield, MN

- Resolved **50+ tickets per week** for **3,000+** users with a **90%+ same-day resolution rate** by diagnosing hardware, software, and account issues across Windows and Mac environments.

AI Dark Patterns Research

Fall 2026

University of St. Thomas, Faculty-Sponsored

Minneapolis, MN

- Conducting **faculty-sponsored research** on AI dark patterns and digital addiction; designing survey instruments and analyzing manipulative design patterns in consumer AI products.

ColorStack

Nov. 2024 – Present

Member

- Engage with a national CS community of **10,000+ members** focused on career development and technical growth for underrepresented students in tech.

Club Soccer

Sep. 2024 – May 2025

Second Captain

- Led a **20+ person roster**, coordinating 2 weekly practices and mentoring teammates to improve team retention.